Does the Fusion Weight Track Query Complexity? Adaptive Score Fusion of Cross-Encoder and Graph-Contextual Evidence for Single- and Multi-hop Passage Ranking

Anonymous Authors
(author information to be added)

Abstract

Graph-augmented rerankers such as ReGAT-BERT score a passage with a linear fusion $\alpha\, s_{sem} + (1-\alpha)\, s_{ctx}$ of a cross-encoder score s_{sem} and a contextual score s_{ctx} produced by a graph attention network (GAT) over the candidate set, with one global α. We test a natural hypothesis: *simple, single-hop queries should rely on the cross-encoder ($\alpha > 0.5$), complex multi-hop queries on the contextual score ($\alpha < 0.5$), so an α that depends on the query should beat a fixed one.* We ablate four score normalisers and five ways of setting α (fixed; a function of the query, of score statistics, or of both; plus explicit complexity features) on MS MARCO and four multi-hop collections (MuSiQue, HotpotQA, BEIR-HotpotQA, 2WikiMultiHopQA), with every upstream model frozen and all selection on validation data. The hypothesis holds in a qualified form. The best fixed α is above 0.5 on MS MARCO and falls below 0.5 on 2WikiMultiHopQA, on MuSiQue's validation split and on its 4-hop test questions; within MuSiQue the test optimum falls with the hop count (0.80, 0.55, 0.40 for 2, 3, 4 hops), and within 2WikiMultiHopQA comparison questions favour the cross-encoder while bridge-comparison and inference questions favour the GAT. Across 17 query buckets, however, the optimum is explained by the relative strength of the two signals (Spearman $\rho = 0.79$ with the CE$-$GAT gap) rather than by complexity as such: on HotpotQA, bridge questions favour the cross-encoder. The second half of the hypothesis is not supported by learned gates: although a per-query oracle α would add 2.3–7.4 nDCG@10 points, the query-dependent gates gain at most 0.04 points over the fixed weight under the primary normaliser (11 of 76 gate/normaliser/dataset cells significantly better, 13 worse), mostly because early stopping returns their constant initialisation. A coarse question-type α chosen on validation data gives small significant gains (up to +0.42).

1 Introduction

Neural rerankers increasingly combine a pointwise relevance model with evidence drawn from the other candidates: graph-based score regularisation (Diaz, 2005; Kurland and Lee, 2005), corpus-graph reranking (MacAvaney et al., 2022b), and graph-attention models over passages and their links (Asai et al., 2020). ReGAT-BERT (reg, n.d.) is one such design. A BERT cross-encoder (Nogueira and Cho, 2019) gives a semantic score $s_{sem}(q,p)$; a GAT (Veličković et al., 2018) refines bi-encoder passage embeddings over a graph of candidates, and the cosine with the query gives a contextual score $s_{ctx}(q,p)$. The final score is

$$s(q,p) = \alpha\, s_{sem}(q,p) + (1-\alpha)\, s_{ctx}(q,p), \quad (1)$$

with a single α for all queries.

A single weight is a strong assumption. A single-hop query ("*what is the capital of Peru*") can be answered by a passage judged in isolation, so the cross-encoder should dominate. A multi-hop question ("*who directed the film whose lead actress was born in Lima*") needs passages that are only relevant through their relation to other passages, which is exactly what a graph over the candidate set can encode. This suggests the hypothesis we test:

> **Hypothesis.** For simple single-hop queries $\alpha > 0.5$ (the semantic score matters more); for complex multi-hop queries $\alpha < 0.5$ (the contextual score matters more). Hence α should depend on the query, and an α conditioned on the query (Q), on the query and the score distributions (QS) or on the score distributions alone (S) should beat a fixed α.

We split the hypothesis into falsifiable claims and test them in a controlled ablation in which

only the fusion step changes (§4). The BM25 candidates, the cross-encoder, the bi-encoder and the GAT are trained or loaded once, frozen and cached, so every fusion variant sees identical inputs. Our contributions are:

- A 4×5 ablation of fusion (normaliser $\times$ what α depends on) on MS MARCO and four multi-hop collections, with oracle upper bounds, graph controls, transfer experiments and pre-registered hypothesis tests (written down before any multi-hop run).
- Evidence that the *optimal* fusion weight does vary with query complexity in the predicted direction on the hardest multi-hop data (MuSiQue, 2WikiMultiHopQA), but that it is governed by the relative reliability of the two signals ($\rho = 0.79$ across buckets), which complexity only partly determines (§5.2, §5.3).
- Evidence that learned query-dependent gates do not realise this variation: they collapse to the fixed weight, and the few significant gains are either tiny or repair a weak normaliser (§5.4). We trace this to early stopping, weak per-query signal and a train/val shift.

2 Related Work

Score fusion and normalisation. Linear combination of retrieval scores goes back to CombSUM and its variants (Fox and Shaw, 1994); because scores live on different scales, per-query normalisation (min–max, z-score) is standard (Montague and Aslam, 2001), and rank-based fusion such as RRF (Cormack et al., 2009) avoids scores altogether. We compare four normalisers because the numerical value of α is only meaningful relative to the scales of s_{sem} and s_{ctx}.

Graph-based and contextual reranking. Score regularisation smooths scores over a document-similarity graph (Diaz, 2005); structural reranking uses graphs induced by language models (Kurland and Lee, 2005); adaptive reranking expands the candidate pool through a corpus graph (MacAvaney et al., 2022b). ReGAT-BERT (reg, n.d.) learns the propagation with a GAT and fuses it with a cross-encoder through a fixed weight, the component we study. We use GATv2 layers (Brody et al., 2022) implemented in PyTorch Geometric (Fey and Lenssen, 2019).

Multi-hop retrieval. Multi-hop QA datasets (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022) require combining several supporting paragraphs; retrievers for them follow hyperlinks or retrieve iteratively (Asai et al., 2020; Xiong et al., 2021). Many HotpotQA questions can be answered with single-hop shortcuts (Min et al., 2019), which motivated MuSiQue's construction (Trivedi et al., 2022); this matters for our results on HotpotQA (§5.3).

3 Hypotheses and Tests

The hypothesis in §1 makes three claims, which we test separately.

C1 (dataset level). On single-hop data the best fixed α exceeds 0.5; on multi-hop data it is below 0.5.

C2 (query level). Within a dataset, the best α falls as complexity rises (more hops; bridge or inference rather than comparison questions).

C3 (method). A query-dependent gate (Q, QS, S) beats the best fixed α, and gains more on multi-hop than on single-hop data.

Before running any multi-hop experiment we also fixed four formal hypotheses and their statistics (Table 8): **H1**, the gate's learned α decreases with complexity (Spearman with hop count on MuSiQue; bridge minus comparison on HotpotQA and BEIR-HotpotQA); **H2**, the adaptive gain is larger than on MS MARCO; **H3**, the oracle headroom is larger than on MS MARCO; **H4**, effects that come from the graph should vanish with a random or empty graph. H1–H3 on the three primary multi-hop datasets form one Holm family (Holm, 1979); everything else is labelled exploratory.

Interpretation rule. α measures the *relative reliability* of the two normalised signals, not "how much context a query needs". We therefore never read raw α without the CE-only and GAT-only effectiveness of the same queries and the normaliser it was obtained with.

4 Experimental Setup

4.1 Frozen pipeline

Candidates. MS MARCO and BEIR-HotpotQA use BM25 (Pyserini, k_1=0.9, b=0.4; Robertson and Zaragoza, 2009; Lin et al., 2021) top-100. MuSiQue, HotpotQA (distractor) and 2Wiki are closed pools of 20 or 10 paragraphs that always contain the gold paragraphs; we order each pool by in-pool BM25, which is also the tie-breaker everywhere. **Semantic score.** s_{sem} is the raw logit

of `cross-encoder/ms-marco-MiniLM-L-6-v2` (Wang et al., 2020), never fine-tuned; on multi-hop data the passage is "*title*. *text*". **Contextual score.** Node features are frozen `msmarco-MiniLM-L6-cos-v5` embeddings (Reimers and Gurevych, 2019). For each query we build a graph over its candidates only (the query is not a node): a symmetrised k-NN graph on cosine similarity (k=8 for K=100, k=3 for closed pools, minimum cosine 0.3) with self-loops. Two GATv2 layers (4×128 heads, then 384) with a residual connection and LayerNorm refine the embeddings, and s_{ctx} is the cosine between the query embedding and the refined passage embedding. One GAT per dataset is trained once on a disjoint *GAT-train* split with a multi-positive listwise InfoNCE loss (τ=0.05; van den Oord et al., 2018), early-stopped on validation, and frozen. A gate checks that GAT-only is not worse than the raw bi-encoder on validation; it passes on every dataset.

4.2 Fusion variants

Normalisers (per query over its candidates): *none*, *min–max*, *z-score*, *rank*. The primary normaliser, fixed on MS MARCO validation before any multi-hop run, is min–max. **What α depends on.** FIXED: a grid search $\alpha \in \{0, 0.05, \ldots, 1\}$ on validation (main baseline, α_0) and a learned scalar. Q: $\alpha = \sigma(\mathrm{MLP}(\mathbf{q}))$ on the query embedding. S: $\alpha = \sigma(\mathrm{MLP}(\mathbf{f}))$ on 12 statistics of the raw score lists (mean, std, max, top-1/top-2 gap and softmax entropy of each score; Spearman correlation and top-m overlap between them). QS: both inputs. QS+C: QS plus query length and spaCy entity count. Each MLP has one hidden layer of 64 units, its output bias is initialised to $\mathrm{logit}(\alpha_0)$ and its last-layer weights near zero, so every gate starts as the fixed baseline. **Training.** Gates are trained on a disjoint *fusion-train* split with a multi-positive listwise softmax loss (Cao et al., 2007) on s/T (learned temperature), plus $\lambda(\alpha - \alpha_0)^2$ with λ=0.01 (and λ=0 as an ablation), AdamW (Loshchilov and Hutter, 2019), early stopping on validation nDCG@10 with the best epoch restored (epoch 0 = the initialisation), 5 seeds. **ReGAT-BERT-style row**: $a\,s_{sem} + b\,s_{ctx}$ with free a, b trained with pointwise BCE, our reproduction of the original training objective. **Oracle**: the per-query α on the grid that maximises that query's test nDCG@10; an upper bound, not a method.

4.3 Data

Table 1 lists the collections. MS MARCO is used under two protocols: **A** trains on MS MARCO train and tests on dev/small; **B** splits dev/small itself, so neither the GAT nor the gate's validation data shares a distribution with the cross-encoder's training data. Complexity labels: MuSiQue hop count (2–4), HotpotQA and BEIR-HotpotQA question type (bridge vs. comparison; BEIR queries joined to HotpotQA by id), 2Wiki question type (comparison, compositional, inference, bridge-comparison), and for MS MARCO only rule-based proxies (length, wh-word, entities). Labels never enter a gate.

4.4 Evaluation

We report nDCG, MRR, Precision and Recall at $k \in \{1, 3, 5, 10, 50\}$, MAP, and AllRecall@$k$ (all gold paragraphs in the top k), computed with ir_measures (MacAvaney et al., 2022a); nDCG@10 is the primary metric (full tables in Appendix A). Every choice (α_0, early stopping, the primary normaliser) is made on validation data; test sets are used once. Differences are tested with a paired bootstrap over queries (10,000 resamples; Efron and Tibshirani, 1993) on seed-averaged per-query scores.

5 Results

5.1 Overall effectiveness

Table 2 shows the headline numbers. Fusion helps little over the cross-encoder alone: the best fixed α adds +0.98 nDCG@10 on 2Wiki, where the GAT is as strong as the cross-encoder, +0.19 on MuSiQue and +0.15 on MS MARCO B, and is *below* CE-only on MS MARCO A (40.89 vs. 41.34) because the α_0 chosen on validation (0.70) is far from the test optimum (0.95; §5.2). The ReGAT-BERT-style free-weight BCE row is significantly worse than the fixed α on MS MARCO and BEIR-HotpotQA (-1.1 and -3.8) and significantly better on MuSiQue (+0.46). The adaptive gates gain at most 0.04 points over the fixed α on any dataset, and QS loses 0.12 on 2Wiki. The oracle rows show what a perfect per-query weight could achieve: +2.3 (2Wiki) to +7.4 (MuSiQue) nDCG@10 points, so the lack of gains is not due to a lack of headroom.

Dataset	Setting	K	GAT train	Fusion train	Val	Test	Gold/pool	Complexity labels
MS MARCO (A)	open (BM25)	100	30,000	10,000	2,000	6,980	–	proxies only
MS MARCO (B)	open (BM25)	100	1,870	1,150	698	1,745	–	proxies only
MuSiQue	closed pool	20	10,000	7,000	2,938	2,417	2.65	hops 2/3/4
HotpotQA	closed pool	10	20,000	10,000	2,000	7,405	2.00	bridge/comparison
BEIR-HotpotQA	open (BM25)	100	20,000	10,000	2,000	7,405	1.59	bridge/comparison
2Wiki	closed pool	10	20,000	10,000	2,000	12,576	2.44	4 types

Table 1: Datasets and splits (queries). Val is always carved from the training data; test is the official dev set (MS MARCO A: dev/small; MuSiQue, HotpotQA, 2Wiki: dev; BEIR: test). MS MARCO B splits dev/small itself, so the cross-encoder never saw any of its queries. Gold/pool = mean number of relevant passages among the K test candidates.

	MSM-A		MSM-B		MuSiQue		HotpotQA		BEIR-HQA		2Wiki	
	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR
BM25 (first stage / in-pool)	22.84	18.40	22.96	18.51	55.11	75.70	81.74	87.87	63.29	80.32	87.77	96.51
SBERT (s_{bi})	35.30	30.38	35.74	30.86	67.08	81.83	82.94	87.19	53.67	70.29	89.19	96.89
CE only (α=1)	41.34	36.39	41.54	36.58	71.09	87.32	89.84	95.42	72.61	89.02	90.77	98.16
GAT only (α=0)	35.42	30.43	35.70	30.58	66.81	81.48	84.14	88.39	55.47	72.63	91.00	97.54
ReGAT-BERT-style (a, b; BCE)	39.83$^{\triangledown}$	34.72$^{\triangledown}$	40.25$^{\triangledown}$	35.22$^{\triangledown}$	71.74$^{\blacktriangle}$	87.29$^{\blacktriangle}$	89.89	95.35	68.81$^{\triangledown}$	85.35$^{\triangledown}$	91.73	98.56
Fixed α (grid on val)	40.89	35.86	41.69	36.84	71.28	86.28	89.87	95.28	72.63	89.03	91.75	98.54
Fixed α (learned)	40.89	35.86	41.69	36.84	71.28	86.28	89.89	95.32	72.63	89.03	91.75	98.54
Adaptive $\alpha(q)$ Q	40.89$^{\blacktriangle}$	35.87	41.70	36.83	71.28	86.28	89.87	95.28	72.63	89.03	91.77	98.37$^{\triangledown}$
Adaptive $\alpha(q, s)$ QS	40.89	35.86	41.70	36.84	71.27	86.28	89.91	95.49	72.63	89.03	91.63$^{\triangledown}$	98.40$^{\triangledown}$
Adaptive $\alpha(s)$ S	40.90	35.87	41.71$^{\blacktriangle}$	36.86	71.27	86.31	89.93	95.49$^{\blacktriangle}$	72.64	89.03	91.75	98.53
Adaptive QS+C	–	–	–	–	71.28	86.28	89.92	95.52$^{\blacktriangle}$	72.63	89.03	91.63$^{\triangledown}$	98.35$^{\triangledown}$
Oracle per-query α (bound)	45.90	41.81	46.01	41.95	78.72	92.50	92.86	97.69	75.56	92.04	94.00	99.49

Table 2: Main results on the test sets (nDCG@10 and MRR@10, $\times 100$, mean over 5 seeds) under the primary normaliser (min–max, fixed on MS MARCO val). $^{\blacktriangle}/^{\triangledown}$: significantly better/worse than the best fixed α (paired bootstrap over queries, 10k resamples, $p < 0.05$, uncorrected). The oracle row picks α per query on the test labels and is an upper bound, not a method. QS+C was added with the multi-hop extension and not run on MS MARCO (–). Full metrics at $k \in \{1, 3, 5, 10, 50\}$ are in Appendix A.

	Test nDCG@10 ($\times 100$)			Best fixed α on val, by normaliser				min–max α^*		
Dataset	CE	GAT	CE−GAT	none	min–max	z-score	rank	val	test	favours
MS MARCO (A)	41.3	35.4	+5.9	0.05	0.70	0.70	0.55	**0.70**	0.95	sem/sem
MS MARCO (B)	41.5	35.7	+5.8	0.15	0.85	0.85	1.00	**0.85**	0.90	sem/sem
MuSiQue	71.1	66.8	+4.3	0.05	0.40	0.45	0.45	**0.40**	0.65	ctx/sem
HotpotQA	89.8	84.1	+5.7	0.10	0.85	0.85	0.90	**0.85**	0.90	sem/sem
BEIR-HotpotQA	72.6	55.5	+17.1	0.35	0.95	0.95	1.00	**0.95**	0.90	sem/sem
2Wiki	90.8	91.0	-0.2	0.05	0.45	0.50	0.55	**0.45**	0.45	ctx/ctx

Table 3: Dataset-level test of the hypothesis. α weights the cross-encoder: fused $= \alpha\, s_{sem} + (1 - \alpha)\, s_{ctx}$. α^* is the arg-max of mean nDCG@10 over the grid $\{0, 0.05, \ldots, 1\}$; the val value (bold) is the one used by the fixed baseline, the test value is for analysis only. "favours": ctx if $\alpha^* < 0.5$, sem if $\alpha^* > 0.5$ (val/test). With raw scores (none) α mostly reflects the scale ratio of the two scores and is not interpretable.

5.2 C1: Is α below 0.5 on multi-hop data?

Table 3 and Figure 1 give the best fixed α per dataset. With raw scores ("none") α only reflects the ratio of score scales (within a query, cross-encoder logits span about 12 units on average, GAT cosines about 0.5) and is near 0 everywhere, so the claim can only be read under a scale-free normaliser. Under min–max and z-score:

- **MS MARCO** favours the cross-encoder: $\alpha^* = 0.70$ (A) and 0.85 (B) on validation, 0.95 and 0.90 on test. This agrees with C1.
- **2Wiki** favours the GAT: $\alpha^* = 0.45$ on validation and test (0.50 under z-score), the only dataset where GAT-only matches CE-only (91.0 vs. 90.8).
- **MuSiQue** favours the GAT on validation ($\alpha^* = 0.40$) but the cross-encoder on test (0.65).
- **HotpotQA** and **BEIR-HotpotQA** favour the cross-encoder strongly (0.85–0.95), contrary to

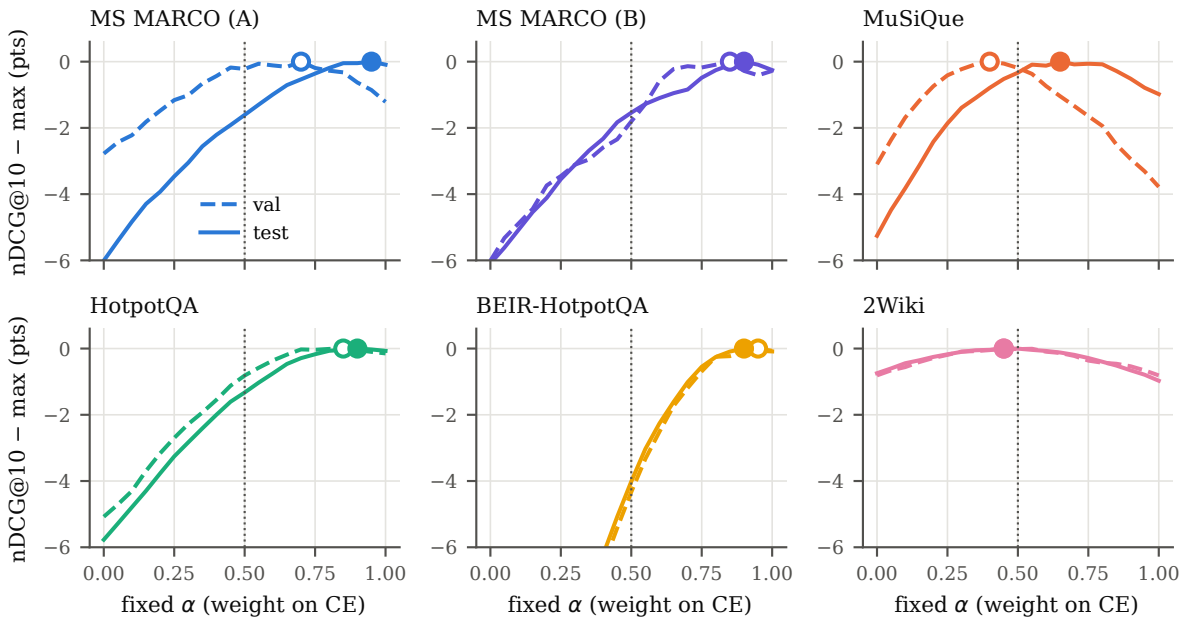


Figure 1: nDCG@10 as a function of the fixed fusion weight α (min–max), relative to each curve's maximum; dashed = validation, solid = test, circles = arg-max. Left of the dotted line the GAT score has more weight. The y-axis is clipped at -6 points.

C1.
The MuSiQue and MS MARCO-A shifts between validation and test have the same cause. Both validation sets are carved from the training questions, i.e. from the same distribution as the GAT's training data, whereas the test sets are the official dev sets. The GAT is relatively stronger on validation (MuSiQue: CE$-$GAT $= -0.7$ points on val vs. $+4.3$ on test; MS MARCO A: $+1.6$ vs. $+5.9$), so validation asks for more GAT weight. Protocol B, whose validation and test both come from dev/small, has no such shift (gap $+5.7$ vs. $+5.8$; α^* 0.85 vs. 0.90). **C1 therefore holds for the two hardest multi-hop collections (2Wiki; MuSiQue in-distribution) and fails for HotpotQA.**

5.3 C2: Does the best α fall with complexity?

Because a learned gate may fail to find structure that exists, we test C2 directly on the data: for every bucket we take the α that maximises the bucket's mean nDCG@10, with a bootstrap CI (Table 4, Figure 2).

MuSiQue supports C2. On test the optimum falls monotonically with the hop count: 0.80 [0.65, 0.80] for 2 hops, 0.55 [0.55, 0.85] for 3 hops and 0.40 [0.25, 0.55] for 4 hops (validation: 0.40, 0.40, 0.25). For 4-hop questions the GAT alone beats the cross-encoder (63.3 vs. 61.3 nDCG@10), and

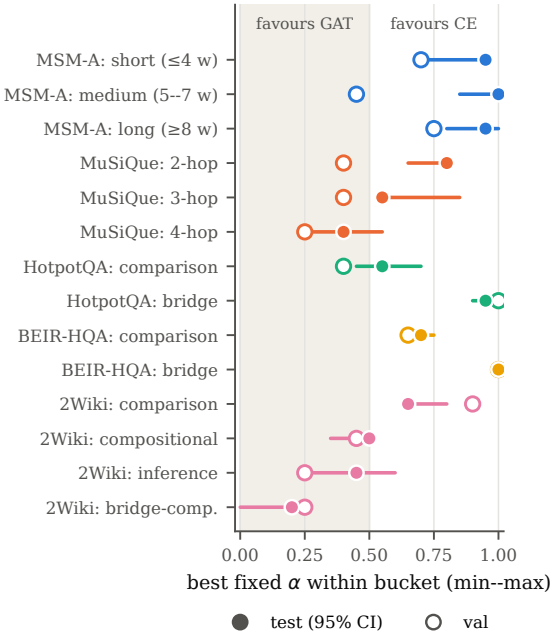


Figure 2: Best fixed α within each complexity bucket (min–max). Filled: test with 95% bootstrap CI; open: validation. Shaded: the GAT score has more weight.

52% of the 4-hop questions whose ranking depends on α prefer the GAT side, against 40% of 2-hop questions.

2Wiki supports C2. Comparison questions, whose two gold paragraphs are each named in

Dataset	Bucket	n (test)	Test nDCG@10 CE	Test nDCG@10 GAT	Best fixed α in bucket val	Best fixed α in bucket test [95% CI]	% queries prefer ctx	Learned $\bar{\alpha}$ (gate)
MS MARCO (A)	short ($\leq$4 w)	2,149	45.0	40.3	0.70	0.95 [0.70, 0.95]	35	–
	medium (5–7 w)	3,291	41.6	35.3	0.45	1.00 [0.85, 1.00]	33	–
	long ($\geq$8 w)	1,540	35.8	28.8	0.75	0.95 [0.80, 1.00]	31	–
MuSiQue	2-hop	1,252	73.8	68.0	0.40	0.80 [0.65, 0.80]	40	0.40
	3-hop	760	71.8	66.7	0.40	0.55 [0.55, 0.85]	39	0.40
	4-hop	405	61.3	63.3	0.25	0.40 [0.25, 0.55]	52	0.40
HotpotQA	comparison	1,487	95.6	94.4	0.40	0.55 [0.45, 0.70]	40	0.90
	bridge	5,918	88.4	81.6	1.00	0.95 [0.90, 1.00]	33	0.88
BEIR-HotpotQA	comparison	1,487	85.7	77.2	0.65	0.70 [0.65, 0.75]	27	0.95
	bridge	5,918	69.3	50.0	1.00	1.00 [1.00, 1.00]	12	0.95
2Wiki	comparison	3,040	97.3	95.5	0.90	0.65 [0.65, 0.80]	30	0.72
	bridge-comp.	2,751	90.1	92.0	0.25	0.20 [0.00, 0.20]	67	0.35
	compositional	5,236	88.5	88.9	0.45	0.50 [0.35, 0.50]	53	0.45
	inference	1,549	86.7	87.6	0.25	0.45 [0.25, 0.60]	59	0.30

Table 4: Query-level test of the hypothesis (min–max). Best fixed α = arg-max of the bucket's mean nDCG@10 curve; CI from 2,000 bootstrap resamples of the bucket's queries. "% prefer ctx": among queries whose nDCG@10 changes with α, the share whose optimal α-interval is centred below 0.5 (tie-robust; the remaining queries are indifferent). Learned $\bar{\alpha}$: mean α of the best (on val) in-domain gate in the bucket. Analysis only: no label enters any gate.

the question, have their optimum at 0.65 (val 0.90); compositional questions at 0.50 (0.45); inference at 0.45 (0.25); bridge-comparison questions, which combine a bridge step with a comparison, at 0.20 [0.00, 0.20] (val 0.25), with 67% of α-sensitive queries preferring the GAT side. The GAT beats the cross-encoder on exactly the three non-comparison types.

HotpotQA and BEIR-HotpotQA contradict C2. Bridge questions, the nominally harder type, have the higher optimum (0.95–1.00), and comparison questions the lower one (0.55 and 0.70). Bridge questions are where the cross-encoder's lead over the GAT is largest (6.8 and 19.3 points); many of them can be solved by matching the question against a single paragraph (Min et al., 2019), while for comparison questions the two gold paragraphs are similar to each other and therefore close in the k-NN graph.

MS MARCO shows no complexity effect: every length bucket has its test optimum at 0.95–1.00, in line with the Step 0 diagnosis (Appendix C), where learned and oracle α did not vary with query length, wh-word or entity count.

One variable explains the optima. Figure 3 plots each bucket's optimum against the effectiveness gap between the two signals on the same queries. The rank correlation is $\rho = 0.79$ over 17 buckets from all six test sets: wherever the GAT is at least as good as the cross-encoder (MuSiQue 4-hop, the three non-comparison 2Wiki types), the optimum is below 0.5, and it rises towards 1 as the cross-encoder pulls ahead. The same holds across graphs (Table 9): on HotpotQA, title-link edges raise GAT-only nDCG@10 from 84.1 to 87.4 and move the fixed optimum from 0.85 to 0.65. Query complexity thus matters for α *through* the relative reliability of the two signals: multi-hop questions favour the contextual score when, and only when, the graph-refined score actually ranks their passages better.

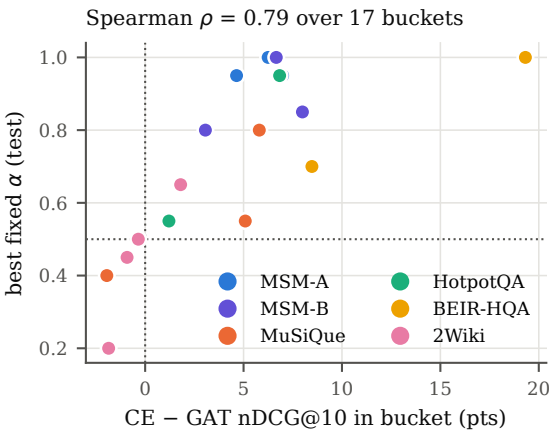


Figure 3: Each point is one complexity bucket (test): the CE−GAT nDCG@10 gap of its queries vs. its best fixed α (min–max).

5.4 C3: Do query-dependent gates beat a fixed α?

Table 5 compares every gate with the best fixed α under the same normaliser on all six test sets. Of 76 comparisons, 11 are significantly positive and 13 significantly negative, and the positives are of two kinds. (i) Gains of at most 0.05 points (MS MARCO under min–max and z-score), which are statistically detectable on 7k queries but practically null. (ii) Larger gains where the fixed baseline of that normaliser is weak: under rank normalisation the gates add up to +1.37 on MuSiQue and +0.41 on 2Wiki, but recover only part of the gap to min–max (MuSiQue: rank 69.58 $\rightarrow$ 70.96 vs. min–max fixed 71.28). The one gain that yields a best-in-dataset system is z-score Q on BEIR-HotpotQA (+0.18, 72.87). There is no sign that gains are larger on multi-hop data (H2, Table 8).

Why the gates do not adapt. Table 6 and Figure 4 show what the gates learn. On MS MARCO, MuSiQue and BEIR-HotpotQA, the standard deviation of the learned α across test queries is at most 0.002: in 100% of these runs the best validation epoch is the initialisation, so early stopping returns the fixed baseline. Training does move the gate (on MuSiQue validation nDCG@10 drops from 78.3 to 75.3 after one epoch for Q), but every move hurts validation, so it is undone. On HotpotQA (QS, S) and 2Wiki (Q, QS) the gates do adapt (sd 0.08–0.24), yet their gains stay within noise. The 2Wiki Q gate learns the right direction (mean α 0.72 for comparison, 0.30 for inference, 0.35 for bridge-comparison questions, cf. Table 4) and gains significantly on bridge-comparison questions (+0.16, $p < 0.001$), but loses on inference questions (-0.23, $p = 0.016$), which cancels the gain overall. On HotpotQA the QS gate puts *higher* α on comparison than on bridge questions (0.90 vs. 0.88), the opposite of what their optima ask for. Three factors explain this: the per-query signal is weak (many queries are indifferent to α: between 14% (MuSiQue) and 59% (MS MARCO B) of test queries have the same nDCG@10 for every α); the listwise training loss on fusion-train does not track validation nDCG@10; and the train/validation shift of §5.2 makes the validation set disagree with the test set. Explicit complexity features (QS+C) and removing the regulariser (λ=0, appendix tables) change nothing.

Coarse, type-aware weights. If gates fail to learn per-query structure, a coarser query-dependent α may still help. Table 7 selects one α per bucket on validation and applies it on test. With gold question types this is significantly better than the global α_0 on HotpotQA (+0.14), BEIR-HotpotQA (+0.42) and 2Wiki (+0.07); hop-count buckets on MuSiQue do not help (-0.03, n.s.) because of the validation/test shift. Query-length buckets, which are available at test time, never help and hurt on MS MARCO A (-0.69), again because validation is unrepresentative there. So a query-dependent α can beat a fixed one when the dependence follows a reliable, coarse property of the query, but the gain is small and depends on labels not available at test time.

5.5 Pre-registered tests

Table 8 reports the nine primary tests exactly as pre-registered. H3 is supported only on MuSiQue: its oracle headroom (7.4 points) exceeds MS MARCO's (5.0) by 2.4 points, whereas HotpotQA and BEIR-HotpotQA have *less* headroom than MS MARCO (3.0 and 2.9 points). H2 is inconclusive everywhere. H1 is "supported" on HotpotQA ($\bar{\alpha}_{\text{bridge}} - \bar{\alpha}_{\text{comp.}} = -0.026$) and "contradicted" on MuSiQue and BEIR-HotpotQA; the latter two verdicts are artefacts, since those gates' α varies by less than 10^{-3}, and the HotpotQA verdict concerns a gate whose α moves in the direction opposite to its own nDCG optimum (§5.4). We therefore base our conclusions on the bucket optima (§5.3), which do not depend on whether a gate learned anything.

5.6 Graph controls (H4)

Table 9 retrains the GAT on five graph types. On MuSiQue the graph structure barely matters: GAT-only nDCG@10 is 67.7 with no edges at all, 66.8 with the default k-NN graph and 68.0 with title links, and the headroom and the (null) adaptive gains are the same for every graph, including random and empty ones. The MuSiQue effects of §5.3 therefore come from the GAT's query–passage representation (trained in-domain with a listwise loss), not from message passing. On HotpotQA, edges matter: title links connect 68% of the gold pairs (vs. 48% for k-NN, 31% for random), raise GAT-only by 3.3 points, lower α_0 to 0.65 and give the best HotpotQA system (90.52). The only significant adaptive gain in this table is on the hybrid graph (+0.18), where 80% of

Norm.	Gate	MSM-A	MSM-B	MuSiQue	HotpotQA	BEIR-HQA	2Wiki
none	Q	-0.48$^{\triangledown}$	+0.01	+0.00	-0.00	-0.00	+0.00
	QS	-0.48$^{\triangledown}$	+0.00	+0.00	+0.01	-0.00$^{\triangledown}$	+0.00
	S	-0.49$^{\triangledown}$	+0.00	+0.02	+0.01	-0.00	-0.00
minmax	Q	**+0.00**$^{\blacktriangle}$	+0.00	+0.00	-0.02	-0.00	+0.03
	QS	+0.00	+0.01	-0.01	+0.02	+0.00	-0.12$^{\triangledown}$
	S	+0.00	**+0.02**$^{\blacktriangle}$	-0.00	+0.04	+0.01	+0.00
	QS+C	–	–	+0.00	+0.03	-0.00	-0.11$^{\triangledown}$
zscore	Q	+0.01	**+0.05**$^{\blacktriangle}$	-0.15$^{\triangledown}$	+0.06	**+0.18**$^{\blacktriangle}$	+0.00
	QS	**+0.02**$^{\blacktriangle}$	-0.01	-0.14$^{\triangledown}$	+0.00	-0.01$^{\triangledown}$	-0.00
	S	+0.01	+0.01	-0.12$^{\triangledown}$	+0.01	-0.01	-0.01$^{\triangledown}$
rank	Q	+0.01	+0.02	**+0.43**$^{\blacktriangle}$	-0.01	-0.01	**+0.30**$^{\blacktriangle}$
	QS	-0.12	-0.04	**+1.22**$^{\blacktriangle}$	-0.04	-0.04$^{\triangledown}$	**+0.41**$^{\blacktriangle}$
	S	-0.10	-0.02	**+1.37**$^{\blacktriangle}$	-0.00	-0.02$^{\triangledown}$	**+0.19**$^{\blacktriangle}$

Table 5: Every adaptive gate against the best fixed α under the same normaliser: test ΔnDCG@10 $\times 100$ (paired bootstrap; $^{\blacktriangle}$ significantly better, $^{\triangledown}$ worse, $p < 0.05$ uncorrected).

		In-domain gate: mean (sd over test queries) of learned α				Restored	Mixed gate
Dataset	α_0	Q	QS	S	QS+C	at init	Q
MS MARCO (A)	0.70	0.700 (0.000)	0.700 (0.000)	0.699 (0.002)	–	100%	0.59 (0.12)
MS MARCO (B)	0.85	0.857 (0.002)	0.853 (0.001)	0.850 (0.001)	–	80%	–
MuSiQue	0.40	0.400 (0.000)	0.400 (0.001)	0.400 (0.002)	0.400 (0.001)	100%	0.54 (0.16)
HotpotQA	0.85	0.850 (0.000)	0.883 (0.210)	0.919 (0.081)	0.909 (0.138)	30%	0.68 (0.15)
BEIR-HotpotQA	0.95	0.950 (0.000)	0.950 (0.000)	0.950 (0.000)	0.950 (0.000)	100%	–
2Wiki	0.45	0.474 (0.237)	0.511 (0.126)	0.449 (0.002)	0.493 (0.157)	55%	–

Table 6: What the gates learn (min–max, $\lambda = 0.01$). The gates are initialised at α_0 and early-stopped on val nDCG@10; "restored at init" is the share of adaptive runs (all gates, 5 seeds) whose best val epoch was the initialisation, i.e. the returned model is the fixed baseline. Mixed gate: one Q gate trained on MS MARCO + MuSiQue + HotpotQA (§5.7), its α on each test set.

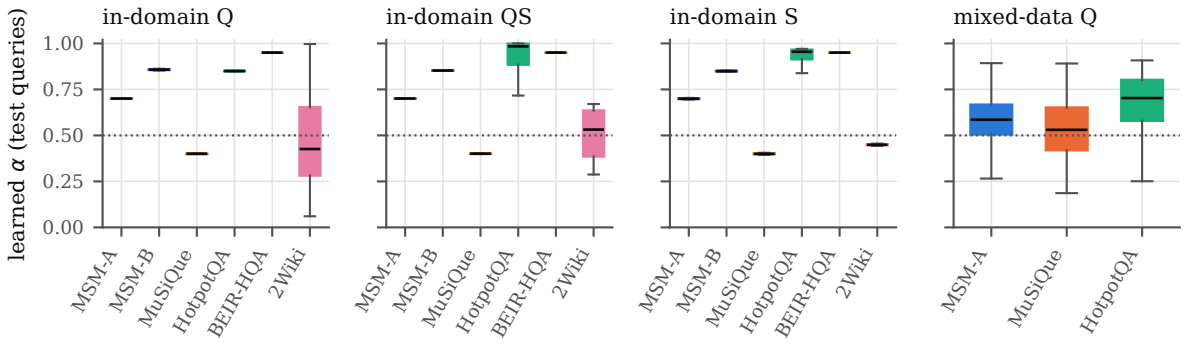


Figure 4: Distribution of the learned α over test queries (seed-averaged; min–max). In-domain gates collapse onto their initialisation α_0 except on HotpotQA (QS, S) and 2Wiki (Q, QS); the gate trained on mixed data (§5.7) spreads α and places MuSiQue lowest.

gold pairs are linked. On MS MARCO the k-NN graph is within 0.1 points of the empty and random graphs (Appendix B).

5.7 Transfer and mixed training

Table 10 tests whether a gate learns to separate simple from complex queries when it sees both. A zero-shot gate trained on MS MARCO stays at its initialisation (sd 0.00); its MuSiQue score is higher than the in-domain gate's only because MS MARCO's $\alpha_0 = 0.70$ happens to be closer to MuSiQue's test optimum (0.65) than MuSiQue's own validation-selected 0.40. A single Q gate trained on MS MARCO, MuSiQue and HotpotQA together does adapt (sd 0.12–0.16) and orders the datasets as the hypothesis predicts: mean α 0.54

Data	Buckets	α_0	Per-bucket α (chosen on val)	Δ	95% CI
MSM-A	length	0.70	long ($\geq$8 w) 0.75; medium (5–7 w) 0.45; short ($\leq$4 w) 0.70	-0.69$^{\triangledown}$	[-0.87, -0.52]
MSM-A	wh-word	0.70	how 0.85; other 0.55; what 0.45; when 0.85; where 0.85; who 0.85; why 0.85	-0.79$^{\triangledown}$	[-1.01, -0.57]
MSM-B	length	0.85	long ($\geq$8 w) 1.00; medium (5–7 w) 0.85; short ($\leq$4 w) 0.75	-0.03	[-0.30, +0.24]
MuSiQue	length	0.40	long ($\geq$8 w) 0.40; medium (5–7 w) 0.15	-0.01	[-0.02, +0.00]
MuSiQue	hops (gold)	0.40	2-hop 0.40; 3-hop 0.40; 4-hop 0.25	-0.03	[-0.14, +0.08]
HotpotQA	length	0.85	long ($\geq$8 w) 0.80; medium (5–7 w) 0.85	-0.03	[-0.09, +0.02]
HotpotQA	type (gold)	0.85	bridge 1.00; comparison 0.40	+0.14$^{\blacktriangle}$	[+0.02, +0.25]
BEIR-HQA	length	0.95	long ($\geq$8 w) 0.95; medium (5–7 w) 1.00	-0.00	[-0.01, +0.00]
BEIR-HQA	type (gold)	0.95	bridge 1.00; comparison 0.65	+0.42$^{\blacktriangle}$	[+0.30, +0.53]
2Wiki	length	0.45	long ($\geq$8 w) 0.45; medium (5–7 w) 0.20	+0.00	[-0.02, +0.02]
2Wiki	type (gold)	0.45	bridge-comp. 0.25; comparison 0.90; compositional 0.45; inference 0.25	+0.07$^{\blacktriangle}$	[+0.01, +0.12]

Table 7: Bucketed fixed α: one α per bucket selected on val and applied on test, against the single global α_0 (test ΔnDCG@10 $\times$100, paired bootstrap). Length buckets are available at test time; gold hop/type labels are not, so those rows measure how much a type-aware α *could* give (analysis only).

Hyp.	Dataset	Gate	Statistic	Value	95% CI	p_{Holm}	Pred.	Verdict
H1	MuSiQue	best=Q	Spearman(α, hops)	+0.107	[+0.067, +0.147]	0.001	<0	contradicted
H2	MuSiQue	best=Q	gain $-$ gain$_{\text{MSM}}$	-0.004	[-0.025, +0.017]	1.000	>0	inconclusive
H3	MuSiQue	oracle	headroom $-$ headroom$_{\text{MSM}}$	+2.435	[+1.942, +2.932]	0.001	>0	supported
H1	HotpotQA	best=QS	$\bar{\alpha}_{\text{bridge}} - \bar{\alpha}_{\text{comp.}}$	-0.026	[-0.037, -0.016]	0.001	<0	supported
H2	HotpotQA	best=QS	gain $-$ gain$_{\text{MSM}}$	+0.011	[-0.095, +0.117]	1.000	>0	inconclusive
H3	HotpotQA	oracle	headroom $-$ headroom$_{\text{MSM}}$	-2.036	[-2.365, -1.710]	0.001	>0	contradicted
H1	BEIR-HotpotQA	best=Q	$\bar{\alpha}_{\text{bridge}} - \bar{\alpha}_{\text{comp.}}$	+0.000	[+0.000, +0.000]	0.001	<0	contradicted
H2	BEIR-HotpotQA	best=Q	gain $-$ gain$_{\text{MSM}}$	-0.005	[-0.025, +0.016]	1.000	>0	inconclusive
H3	BEIR-HotpotQA	oracle	headroom $-$ headroom$_{\text{MSM}}$	-2.078	[-2.427, -1.739]	0.001	>0	contradicted

Table 8: Pre-registered primary tests (written before any multi-hop run; Holm over the nine tests). H1 uses the best (on val) gate; H2/H3 values are nDCG@10 $\times$100. H1 verdicts on MuSiQue and BEIR are computed on gates whose α varies by less than 10^{-3} across queries (Table 6), so their sign is not interpretable.

Dataset	Graph	Degree	Gold pairs linked (%)	GAT nDCG@10	α_0	Fixed nDCG@10	Δ adaptive (best gate)	Headroom (oracle)
MuSiQue	none (self-loops)	0.00	0	67.7	0.35	71.75	-0.03	7.7
	random (same degrees)	2.78	14	67.0	0.50	71.89	-0.03	6.6
	kNN emb. (default)	2.78	27	66.8	0.40	71.28	+0.00	7.4
	title link	0.81	19	68.0	0.35	71.97	+0.00	7.6
	kNN $\cup$ title	3.20	36	67.0	0.40	71.40	+0.00	7.4
HotpotQA	none (self-loops)	0.00	0	84.6	0.85	89.98	+0.02	3.0
	random (same degrees)	2.78	31	83.7	0.90	89.89	-0.02	2.8
	kNN emb. (default)	2.78	48	84.1	0.85	89.87	+0.02	3.0
	title link	0.98	68	87.4	0.65	90.52	+0.05	3.5
	kNN $\cup$ title	3.18	80	85.5	0.65	89.97	+0.18$^{\blacktriangle}$	3.4

Table 9: Graph controls (H4, exploratory): a GAT retrained per graph type, everything else fixed. Degree excludes self-loops; "gold pairs linked" = share of supporting-paragraph pairs joined by an edge (test). nDCG@10 and Δ are $\times$100 on test; Δ = best gate (on val) $-$ best fixed α, $^{\blacktriangle}$ $p < 0.05$.

on MuSiQue, 0.59 on MS MARCO and 0.68 on HotpotQA. It still loses to the in-domain fixed weights on MS MARCO (-0.60) and HotpotQA (-0.45), so the gate captures the coarse dataset-level difference but not enough of it to pay off.

			Transferred gate, test nDCG@10				In-domain		Q gate
Setting	Train	Test	fixed	Q	QS	S	fixed	α_0	$\bar{\alpha}$ (sd)
leave-one-out	MSM+HQA	MuSiQue	71.99	71.93	71.98	72.00	71.28	0.70	0.73 (0.02)
mixed	MSM+MuSiQue+HQA	MS MARCO (A)	40.13	40.29	40.18	40.22	40.89	0.55	0.59 (0.12)
mixed	MSM+MuSiQue+HQA	HotpotQA	88.88	89.44	89.32	89.34	89.89	0.55	0.68 (0.15)
mixed	MSM+MuSiQue+HQA	MuSiQue	71.98	71.77	71.81	71.79	71.28	0.55	0.54 (0.16)
zero-shot	MSM	BEIR-HotpotQA	71.45	71.45	71.45	71.46	72.63	0.70	0.70 (0.00)
zero-shot	MSM	HotpotQA	89.24	89.24	89.25	89.20	89.89	0.70	0.70 (0.00)
zero-shot	MSM	MuSiQue	71.78	71.78	71.80	71.79	71.28	0.70	0.70 (0.00)

Table 10: Transfer and mixed training of the gate (E4, min–max, $\times 100$, 5 seeds). Zero-shot uses the frozen MS MARCO GAT for s_{ctx}; mixed and leave-one-out use each dataset's own GAT. α_0 is the fixed α the gate starts from (selected on the training sources' val).

Claim	Evidence	Verdict
C1: $\alpha > 0.5$ single-hop	MS MARCO 0.70–0.95	supported
C1: $\alpha < 0.5$ multi-hop	2Wiki 0.45; MuSiQue 0.40 (val), 0.65 (test); HotpotQA 0.85–0.95	partly
C2: α falls with complexity	MuSiQue 0.80/0.55/0.40; 2Wiki comp. 0.65 vs. others 0.20–0.50; HotpotQA reversed	partly
C3: adaptive > fixed	11/76 cells better, 13 worse; gains $\leq$0.04 under primary norm.; oracle 2.3–7.4	not supported
C3: larger gain on multi-hop	H2 inconclusive on all datasets	not supported

Table 11: Verdict on the hypothesis.

6 Discussion

Table 11 summarises the evidence. The intuition behind the hypothesis is right in one important sense: the fusion weight that a query *should* get is not constant, it differs by up to 0.65 between question types of one dataset (2Wiki validation: 0.90 vs. 0.25), and on the hardest multi-hop questions it moves to the contextual side. What the hypothesis gets wrong is the mechanism and the method. The mechanism is relative reliability: complexity lowers α^* only where it makes the cross-encoder, which reads each passage in isolation, worse than the graph-refined score, as on 4-hop MuSiQue and bridge-comparison 2Wiki questions; where single-paragraph matching still works (HotpotQA bridge questions) the cross-encoder keeps the weight. The method fails because a gate must infer, from one query embedding or a dozen score statistics, a property that changes the optimal weight for only a minority of queries, and early stopping on a validation set that is itself shifted rejects every move.

Two practical lessons follow. First, the fixed weight should be selected on validation data drawn from the test distribution: on MS MARCO A and MuSiQue the test-optimal fixed α is 0.25 above the validation choice and would add 0.54 and 0.79 nDCG@10 points, more than any gate. Second, improving the contextual signal itself (title links on HotpotQA: +0.65 points for fixed fusion) is worth more than making its weight adaptive.

7 Conclusion

We tested whether the weight between a cross-encoder and a graph-contextual score should depend on query complexity. The optimal weight does vary: it is above 0.5 on MS MARCO, below 0.5 on 2WikiMultiHopQA, and it falls with the hop count on MuSiQue. This variation is explained by the relative reliability of the two scores ($\rho = 0.79$ across buckets) rather than by complexity as such. Learned query-dependent gates, however, do not exploit it. They collapse to the fixed weight and gain at most 0.04 nDCG@10 points over it under the primary normaliser, although an oracle shows 2.3–7.4 points of headroom. A coarse weight per question type is the only query-dependent scheme that helps, and only modestly.

Limitations

All models are small MiniLM checkpoints trained on MS MARCO and are never fine-tuned on the multi-hop data; a cross-encoder trained on multi-hop data would shift every optimum towards $\alpha = 1$. Three of the multi-hop settings are closed pools in which the gold paragraphs are always present, so they test reranking, not retrieval. The graphs are per-query candidate graphs; ReGAT-

BERT's original structural edges cannot be built on MS MARCO, and our results concern this adaptation, not a replication. The validation sets of MS MARCO A and the closed-pool datasets are carved from training questions, which causes the validation/test shift we report. Complexity labels on MS MARCO are heuristic proxies. Bucket optima are computed on test data for analysis; they are never used to select a reported system, but the type-aware results in Table 7 use gold labels unavailable at test time. Significance marks in the ablation grid are uncorrected; only the nine primary tests are Holm-corrected.

Ethics Statement

We use public research datasets under their licences and release no new data. The work does not involve human subjects or personal data beyond what these benchmarks already contain.

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A Full results

Tables 12–17 report nDCG, MRR, Precision and Recall at $k \in \{1, 3, 5, 10, 50\}$ and MAP on every test set for the reference rows, all gates (with $\lambda = 0.01$ and $\lambda = 0$), the BCE loss ablation and the oracle. Precision@k divides by k; on closed pools the pool size bounds Recall@50 and Precision@50. Table 18 gives the out-of-distribution TREC DL 2019/2020 results (Craswell et al., 2020, 2021) for MS MARCO.

B GAT graph sensitivity

Table 19 gives the validation effectiveness of s_{ctx} alone for the graph controls on MS MARCO and the k sweep on the closed pools. The default (k=8, 0.3 on MS MARCO; k=3 on closed pools) was fixed in advance and not changed after the sweep.

C Step 0: MS MARCO diagnosis

Before adding multi-hop data we checked whether "$\alpha > 0.5$ because MS MARCO queries are simple" explains the MS MARCO result. CE-only beats GAT-only by only 1.6 nDCG@10 points on validation; the best fixed α moves from 0.05 (none) to 0.70 (min–max, z-score) and 0.55 (rank), so its value is partly a scale artefact; the learned α hardly varies (sd 0.0003–0.006); and its Spearman correlation with query length and entity count is slightly *positive* (+0.02 to +0.08), the opposite of the prediction (Figure 5). We recorded the explanation as not supported on MS MARCO and fixed min–max as the primary normaliser.

D Reproducibility

All numbers are produced by the released pipeline (`python -m src.pipeline`); this report is generated from its result files by `python -m src.report_analysis` and `python -m src.report_paper`. Software: PyTorch 2.15, PyTorch Geometric 2.8.1, sentence-transformers 6.2, Pyserini 1.6 (Lucene BM25 indexes `msmarco-v1-passage` and `beir-v1.0.0-hotpotqa.flat`), ir_datasets 0.6.3, ir_measures 0.4.3, spaCy 3.8 with `en_core_web_sm`. Hardware: one NVIDIA RTX 6000 Ada (48 GB). The complete multi-hop extension (golden regression of MS MARCO, four datasets, graph controls, transfer, hypothesis tests) ran in 1 h 22 min; the MS MARCO study in under an hour. A golden regression confirmed that the variable-length (masked) code reproduces all 2×182 MS MARCO runs exactly (maximum per-query difference 3×10^{-8}). Deviations from the original protocol (e.g. the per-query k-NN graph instead of ReGAT-BERT's structural graph, padding of short BM25 runs, a learnable bias in the BCE row) are logged in the repository's `NOTES.md`.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	10.5	17.0	19.7	22.8	27.4	10.5	15.6	17.1	18.4	19.4	10.5	7.5	5.9	3.9	1.2	10.2	21.7	28.5	37.9	58.1	19.2
SBERT (s_{bi})	20.4	29.3	32.4	35.3	38.2	20.4	27.5	29.2	30.4	31.1	20.4	12.3	9.0	5.4	1.4	19.8	35.7	43.2	52.0	64.4	30.6
CE only (α=1)	25.5	35.6	38.8	41.3	43.1	25.5	33.6	35.4	36.4	36.8	25.5	14.8	10.5	6.1	1.4	24.7	42.9	50.5	58.2	65.4	36.2
GAT only (α=0)	20.4	29.2	32.4	35.4	38.2	20.4	27.4	29.2	30.4	31.1	20.4	12.3	9.0	5.5	1.4	19.8	35.6	43.3	52.4	64.5	30.6
ReGAT-BERT-style (a, b; BCE)	23.9	33.8	37.1	39.8	41.8	23.9	31.8	33.6	34.7	35.2	23.9	14.2	10.2	6.0	1.4	23.2	41.0	48.8	57.1	65.4	34.6
Fixed α (grid on val)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Fixed α (learned)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(q)$ Q	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(q, s)$ QS	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(s)$ S	25.0	35.1	38.4	40.9	42.7	25.0	33.1	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.2	57.9	65.4	35.7
Q, $\lambda = 0$	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
QS, $\lambda = 0$	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
S, $\lambda = 0$	25.0	35.1	38.4	40.9	42.7	25.0	33.1	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.2	57.9	65.4	35.7
Fixed α, BCE loss (minmax)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
QS, BCE loss (minmax)	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Oracle per-query α (bound)	32.1	41.3	43.9	45.9	47.1	32.1	39.6	41.0	41.8	42.1	32.1	16.6	11.3	6.3	1.4	31.2	47.9	54.1	60.1	65.4	41.4

Table 12: MS MARCO (A): all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	10.3	17.4	19.9	23.0	27.6	10.3	15.8	17.2	18.5	19.6	10.3	7.7	5.9	3.9	1.2	10.1	22.5	28.7	37.9	58.1	19.4
SBERT (s_{bi})	20.7	30.2	33.1	35.7	38.6	20.7	28.1	29.7	30.9	31.5	20.7	12.7	9.1	5.4	1.4	20.3	37.0	44.0	51.9	64.2	31.1
CE only (α=1)	25.4	36.2	39.1	41.5	43.3	25.4	33.9	35.6	36.6	37.0	25.4	15.2	10.6	6.1	1.4	24.8	43.9	50.9	58.1	65.3	36.5
GAT only (α=0)	19.9	30.1	33.0	35.7	38.4	19.9	27.9	29.4	30.6	31.2	19.9	12.9	9.2	5.5	1.4	19.5	37.6	44.6	52.7	64.6	30.7
ReGAT-BERT-style (a, b; BCE)	24.5	34.3	37.9	40.2	42.2	24.5	32.2	34.2	35.2	35.7	24.5	14.3	10.4	6.0	1.4	23.9	41.5	50.0	57.1	65.2	35.2
Fixed α (grid on val)	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.2	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Fixed α (learned)	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.2	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Adaptive $\alpha(q)$ Q	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.8	36.8	37.2	26.0	15.2	10.6	6.1	1.4	25.3	43.8	50.9	58.1	65.3	36.7
Adaptive $\alpha(q, s)$ QS	26.0	36.3	39.4	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Adaptive $\alpha(s)$ S	26.1	36.3	39.4	41.7	43.5	26.1	34.2	35.9	36.9	37.3	26.1	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Q, $\lambda = 0$	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.8	36.8	37.2	26.0	15.2	10.6	6.1	1.4	25.3	43.8	50.9	58.1	65.3	36.7
QS, $\lambda = 0$	26.0	36.3	39.4	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
S, $\lambda = 0$	26.1	36.3	39.4	41.7	43.5	26.1	34.2	35.9	36.9	37.3	26.1	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Fixed α, BCE loss (zscore)	25.8	36.4	39.3	41.5	43.4	25.8	34.1	35.7	36.7	37.1	25.8	15.2	10.6	6.0	1.4	25.2	44.0	51.0	57.8	65.3	36.6
QS, BCE loss (zscore)	25.9	36.3	39.3	41.6	43.4	25.9	34.1	35.8	36.7	37.1	25.9	15.2	10.6	6.0	1.4	25.3	43.9	51.0	57.8	65.3	36.6
Oracle per-query α (bound)	32.0	41.8	44.3	46.0	47.3	32.0	39.9	41.2	41.9	42.2	32.0	16.9	11.5	6.3	1.4	31.1	48.8	54.9	60.1	65.4	41.5

Table 13: MS MARCO (B): all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	66.9	48.6	50.9	55.1	68.6	66.9	73.7	74.8	75.7	76.1	66.9	35.5	24.7	14.9	5.3	27.3	42.3	48.6	58.2	100.0	48.9
SBERT (s_{bi})	72.6	54.3	59.4	67.1	74.8	72.6	80.0	81.4	81.8	81.9	72.6	40.7	30.9	20.2	5.3	29.3	47.9	59.9	77.4	100.0	57.2
CE only (α=1)	79.8	61.9	65.4	71.1	78.7	79.8	86.3	87.0	87.3	87.4	79.8	46.5	32.9	20.1	5.3	32.4	55.3	64.3	77.4	100.0	62.6
GAT only (α=0)	72.4	53.9	58.9	66.8	74.7	72.4	79.5	80.9	81.5	81.6	72.4	40.2	30.6	20.2	5.3	29.3	47.4	59.1	77.1	100.0	57.1
ReGAT-BERT-style (a, b; BCE)	79.8	61.7	65.4	71.7	79.0	79.8	86.2	86.9	87.3	87.3	79.8	46.4	33.2	20.6	5.3	32.5	54.7	64.5	78.8	100.0	63.0
Fixed α (grid on val)	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Fixed α (learned)	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(q)$ Q	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(q, s)$ QS	78.3	60.6	64.7	71.3	78.4	78.3	85.1	85.9	86.3	86.3	78.3	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(s)$ S	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive QS+C	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Q, $\lambda = 0$	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
QS, $\lambda = 0$	78.3	60.6	64.7	71.3	78.4	78.3	85.1	85.9	86.3	86.3	78.3	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
S, $\lambda = 0$	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Fixed α, BCE loss (zscore)	79.3	61.0	65.0	71.5	78.6	79.3	85.7	86.5	86.9	86.9	79.3	45.9	33.1	20.6	5.3	32.2	54.1	64.3	79.0	100.0	62.6
QS, BCE loss (zscore)	80.3	61.4	65.2	71.8	78.9	80.3	86.2	87.0	87.4	87.4	80.3	46.1	33.0	20.7	5.3	32.6	54.5	64.2	79.3	100.0	62.9
Oracle per-query α (bound)	87.9	68.5	72.0	78.7	83.5	87.9	91.9	92.3	92.5	92.5	87.9	51.8	36.4	22.5	5.3	35.6	61.1	70.7	85.8	100.0	69.7

Table 14: MuSiQue: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	81.2	67.6	71.9	81.7	81.7	81.2	86.6	87.4	87.9	87.9	81.2	43.5	29.4	20.0	4.0	40.6	65.2	73.6	100.0	100.0	69.3
SBERT (s_{bi})	80.1	69.2	74.7	82.9	82.9	80.1	85.8	86.7	87.2	87.2	80.1	45.2	31.5	20.0	4.0	40.0	67.8	78.8	100.0	100.0	71.7
CE only (α=1)	92.2	81.0	84.7	89.8	89.8	92.2	95.1	95.3	95.4	95.4	92.2	52.9	34.7	20.0	4.0	46.1	79.4	86.7	100.0	100.0	81.8
GAT only (α=0)	81.7	71.1	76.7	84.1	84.1	81.7	87.2	88.0	88.4	88.4	81.7	46.6	32.5	20.0	4.0	40.8	69.9	81.1	100.0	100.0	73.5
ReGAT-BERT-style (a, b; BCE)	92.1	81.1	84.8	89.9	89.9	92.1	95.0	95.3	95.3	95.3	92.1	52.9	34.7	20.0	4.0	46.1	79.4	86.7	100.0	100.0	81.9
Fixed α (grid on val)	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
Fixed α (learned)	92.1	81.1	84.8	89.9	89.9	92.1	95.0	95.2	95.3	95.3	92.1	53.0	34.7	20.0	4.0	46.0	79.5	86.7	100.0	100.0	82.0
Adaptive $\alpha(q)$ Q	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
Adaptive $\alpha(q, s)$ QS	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.4	86.8	100.0	100.0	81.9
Adaptive $\alpha(s)$ S	92.3	81.2	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.5	86.8	100.0	100.0	82.0
Adaptive QS+C	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.5	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.4	86.8	100.0	100.0	81.9
Q, $\lambda = 0$	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
QS, $\lambda = 0$	92.4	81.1	84.9	89.9	89.9	92.4	95.2	95.5	95.5	95.5	92.4	53.0	34.7	20.0	4.0	46.2	79.5	86.8	100.0	100.0	81.9
S, $\lambda = 0$	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.1	79.5	86.8	100.0	100.0	81.9
Fixed α, BCE loss (none)	91.9	81.0	84.8	89.8	89.8	91.9	94.9	95.2	95.2	95.2	91.9	53.0	34.8	20.0	4.0	46.0	79.4	86.9	100.0	100.0	81.9
QS, BCE loss (none)	91.9	81.0	84.8	89.9	89.9	91.9	94.9	95.2	95.3	95.3	91.9	53.0	34.8	20.0	4.0	46.0	79.4	86.9	100.0	100.0	81.9
Oracle per-query α (bound)	96.0	86.1	89.4	92.9	92.9	96.0	97.5	97.7	97.7	97.7	96.0	56.4	36.5	20.0	4.0	48.0	84.5	91.1	100.0	100.0	86.8

Table 15: HotpotQA: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	73.8	58.4	61.1	63.3	66.2	73.8	79.1	79.9	80.3	80.6	73.8	36.6	24.1	13.2	3.0	36.9	54.9	60.3	65.8	76.3	55.0
SBERT (s_{bi})	62.7	48.4	51.1	53.7	57.8	62.7	68.6	69.7	70.3	70.8	62.7	30.1	20.2	11.4	2.9	31.4	45.1	50.4	57.0	72.2	45.4
CE only (α=1)	84.6	68.6	71.0	72.6	74.0	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.9	78.8	64.8
GAT only (α=0)	65.7	50.6	53.0	55.5	59.4	65.7	71.2	72.0	72.6	73.0	65.7	31.4	20.7	11.6	2.9	32.9	47.1	51.8	58.2	72.6	47.3
ReGAT-BERT-style (a, b; BCE)	79.9	64.1	66.7	68.8	70.9	79.9	84.5	85.1	85.4	85.5	79.9	40.4	26.3	14.2	3.1	40.0	60.6	65.8	71.0	78.2	60.6
Fixed α (grid on val)	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Fixed α (learned)	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(q)$ Q	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(q, s)$ QS	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(s)$ S	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive QS+C	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Q, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
QS, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
S, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Fixed α, BCE loss (none)	84.8	68.7	71.1	72.7	74.1	84.8	88.5	89.0	89.1	89.2	84.8	43.4	28.0	14.8	3.1	42.4	65.1	69.9	73.9	78.8	64.9
QS, BCE loss (none)	84.8	68.7	71.1	72.7	74.1	84.8	88.5	89.0	89.1	89.2	84.8	43.4	28.0	14.8	3.1	42.4	65.1	69.9	73.9	78.8	64.9
Oracle per-query α (bound)	89.3	72.3	74.3	75.6	76.5	89.3	91.6	91.9	92.0	92.1	89.3	45.5	28.8	15.1	3.1	44.6	68.3	72.1	75.3	78.8	68.4

Table 16: BEIR-HotpotQA: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	93.8	75.9	78.1	87.8	87.8	93.8	96.4	96.5	96.5	96.5	93.8	52.7	36.1	24.4	4.9	41.6	67.9	77.1	100.0	100.0	76.5
SBERT (s_{bi})	94.4	78.1	80.9	89.2	89.2	94.4	96.8	96.9	96.9	96.9	94.4	54.8	38.3	24.4	4.9	41.9	70.5	80.7	100.0	100.0	79.1
CE only (α=1)	96.6	81.1	83.0	90.8	90.8	96.6	98.1	98.2	98.2	98.2	96.6	56.8	38.6	24.4	4.9	42.9	73.4	82.3	100.0	100.0	81.6
GAT only (α=0)	95.5	81.6	84.7	91.0	91.0	95.5	97.5	97.5	97.5	97.5	95.5	58.1	40.8	24.4	4.9	42.5	74.5	85.5	100.0	100.0	82.4
ReGAT-BERT-style (a, b; BCE)	97.3	83.0	85.0	91.7	91.7	97.3	98.5	98.6	98.6	98.6	97.3	58.6	40.1	24.4	4.9	43.3	75.5	84.7	100.0	100.0	83.4
Fixed α (grid on val)	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Fixed α (learned)	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Adaptive $\alpha(q)$ Q	97.0	83.0	85.4	91.8	91.8	97.0	98.3	98.4	98.4	98.4	97.0	58.8	40.5	24.4	4.9	43.1	75.7	85.4	100.0	100.0	83.6
Adaptive $\alpha(q, s)$ QS	97.0	82.6	85.0	91.6	91.6	97.0	98.4	98.4	98.4	98.4	97.0	58.3	40.2	24.4	4.9	43.2	75.2	84.9	100.0	100.0	83.3
Adaptive $\alpha(s)$ S	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Adaptive QS+C	97.0	82.7	85.1	91.6	91.6	97.0	98.3	98.4	98.4	98.4	97.0	58.4	40.3	24.4	4.9	43.1	75.3	85.0	100.0	100.0	83.3
Q, $\lambda = 0$	96.9	83.0	85.4	91.8	91.8	96.9	98.3	98.3	98.3	98.3	96.9	58.8	40.6	24.4	4.9	43.1	75.7	85.5	100.0	100.0	83.6
QS, $\lambda = 0$	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
S, $\lambda = 0$	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Fixed α, BCE loss (rank)	96.6	82.0	84.4	91.3	91.3	96.6	98.2	98.2	98.2	98.2	96.6	57.7	39.8	24.4	4.9	43.0	74.6	84.3	100.0	100.0	82.6
QS, BCE loss (rank)	96.5	82.9	85.7	91.7	91.7	96.5	98.1	98.1	98.1	98.1	96.5	59.0	41.1	24.4	4.9	42.9	75.8	86.2	100.0	100.0	83.6
Oracle per-query α (bound)	99.0	87.3	89.3	94.0	94.0	99.0	99.5	99.5	99.5	99.5	99.0	62.5	42.6	24.4	4.9	44.1	80.3	89.5	100.0	100.0	87.6

Table 17: 2Wiki: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

		TREC DL 2019			TREC DL 2020		
		nDCG@10	MRR@10	MAP	nDCG@10	MRR@10	MAP
MS MARCO (A)	BM25 (first stage / in-pool)	50.6	70.2	24.8	48.0	65.3	26.9
	SBERT (s_{bi})	66.9	80.9	31.1	60.9	74.5	35.1
	CE only (α=1)	72.7	87.5	35.7	67.6	84.6	39.6
	GAT only (α=0)	67.3	80.4	31.4	59.4	71.8	34.5
	ReGAT-BERT-style (a, b; BCE)	72.2	85.3	34.7	65.3	83.0	38.7
	Fixed α (grid on val)	72.2	86.1	35.3	67.1	83.9	39.2
	Fixed α (learned)	72.2	86.1	35.3	67.1	83.9	39.2
	Adaptive $\alpha(q)$ Q	72.2	86.1	35.3	67.1	83.9	39.2
	Adaptive $\alpha(q, s)$ QS	72.2	85.9	35.3	67.1	83.9	39.2
	Adaptive $\alpha(s)$ S	72.3	86.1	35.3	67.1	83.9	39.2
	Oracle per-query α (bound)	74.8	89.5	35.8	69.9	86.4	40.5
MS MARCO (B)	BM25 (first stage / in-pool)	50.6	70.2	24.8	48.0	65.3	26.9
	SBERT (s_{bi})	66.9	80.9	31.1	60.9	74.5	35.1
	CE only (α=1)	72.7	87.5	35.7	67.6	84.6	39.6
	GAT only (α=0)	66.0	79.5	31.0	59.8	75.7	35.1
	ReGAT-BERT-style (a, b; BCE)	71.7	84.9	34.8	65.3	83.7	38.5
	Fixed α (grid on val)	72.3	87.1	35.7	67.2	83.9	39.6
	Fixed α (learned)	72.3	87.1	35.7	67.2	83.9	39.6
	Adaptive $\alpha(q)$ Q	72.5	87.1	35.7	67.2	83.9	39.5
	Adaptive $\alpha(q, s)$ QS	72.5	87.1	35.7	67.2	83.9	39.6
	Adaptive $\alpha(s)$ S	72.4	87.1	35.7	67.2	83.8	39.5
	Oracle per-query α (bound)	74.8	89.5	35.9	70.2	87.2	40.5

Table 18: Out-of-distribution TREC DL (graded nDCG; MRR and MAP with rel $\geq$ 2), $\times$100. Not used for any decision.

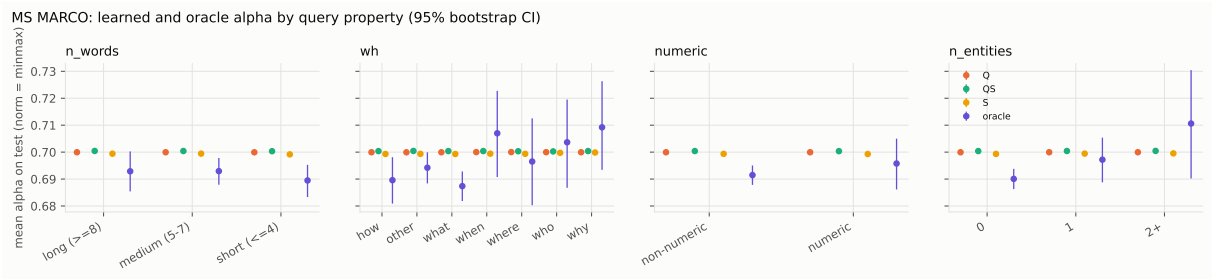


Figure 5: MS MARCO (A, test, min–max): mean learned α of the three gates and mean oracle α by query property, with 95% bootstrap CIs. Learned α is constant; oracle α shows no trend in the predicted direction.

Data	Graph	Deg.	GAT	s_{bi}
MS MARCO (A)	no_graph	1.0	37.88	37.71
MS MARCO (A)	random_graph	12.3	37.94	37.71
MS MARCO (A)	knn_k4_s0.0	7.1	37.73	37.71
MS MARCO (A)	knn_k4_s0.3	6.9	37.76	37.71
MS MARCO (A)	knn_k4_s0.5	5.6	37.65	37.71
MS MARCO (A)	knn_k8_s0.0	13.2	38.00	37.71
MS MARCO (A)	knn_k8_s0.3	12.3	37.98	37.71
MS MARCO (A)	knn_k8_s0.5	9.2	37.90	37.71
MS MARCO (A)	knn_k16_s0.0	24.6	38.04	37.71
MS MARCO (A)	knn_k16_s0.3	22.0	37.92	37.71
MS MARCO (A)	knn_k16_s0.5	14.7	37.76	37.71
MuSiQue	knn_k2	2.0	75.62	70.08
MuSiQue	knn_k3	2.6	75.19	70.08
MuSiQue	knn_k5	3.4	75.27	70.08
HotpotQA	knn_k2	2.2	84.26	82.67
HotpotQA	knn_k3	2.8	84.09	82.67
HotpotQA	knn_k5	3.5	83.93	82.67

Table 19: GAT graph sensitivity on val (nDCG@10 $\times$100 of s_{ctx} alone). knn_kk_sm: k neighbours, min cosine m. MS MARCO degrees include the self-loop.